Software Test Plan (STP) — Task Manager Application

Project: Task Manager Application (SE Mini-Project)
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Task Manager Application v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers registration & authentication, task CRUD, task assignment & the Kanban board, notifications & comments, and search/filter/reporting.
Hosting/infrastructure management is excluded.

References: SRS_TaskManagerApplication.docx v1.3 (TM-F-001–042, TM-NF-001–005, PRJ-SR-001–005), Software Architecture and Design Specification (SAD) v1.0.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), RTM (Requirements Traceability Matrix), JWT (JSON Web Token), RBAC (Role-Based Access Control), SPA (Single-Page Application).

2. Test Items

- Web Client Component (React SPA)
- Auth & User Mgmt Service
- Task & Project Service
- Notification & Collaboration Service
- Search & Reporting Service
- MongoDB Database Component

3. Features to be Tested

Features mapped to the SRS requirement IDs:
- TM-F-001..003: User registration, login, and profile management
- TM-F-010..012: Task creation, editing, and status updates
- TM-F-020..023: Project/member management, task assignment, Kanban board
- TM-F-030..032: In-app notifications and task comments
- TM-F-040..042: Search, filters, dashboard reporting, CSV export

- TM-NF-001: API response time ≤2s for 95% of requests
- TM-NF-002: 99% monthly uptime
- TM-NF-005: 100 concurrent users without noticeable degradation
- PRJ-SR-001..005: Password hashing, JWT expiry, RBAC, TLS, input sanitization

4. Features Not to be Tested

- Underlying cloud hosting infrastructure (assumed managed)
- Third-party email service internals (Nodemailer transport, if used)
- Browser-vendor-specific rendering bugs outside Chrome/Firefox/Edge
- Offline usage (explicitly out of scope per SRS §9)

5. Test Approach / Strategy

Levels:
- Unit tests (per service, e.g. password hashing in Auth & User Mgmt Service)
- Integration tests (Presentation ↔ Business ↔ Data layer interfaces: AuthAPI, TaskBoardAPI, NotificationAPI, SearchReportAPI, AssignmentEvent, and the four *DataAccess interfaces)
- System tests (end-to-end, one per SAD sequence-diagram flow: create & assign a task; user registration & login)
- Acceptance tests (UAT against the exact acceptance criteria already defined per requirement in the SRS's RTM)

Types:
- Functional testing (all SRS feature areas, §4.1–§4.5)
- Regression testing
- Performance testing (TM-NF-001 API latency, TM-NF-005 100 concurrent users)
- Reliability testing (TM-NF-002 uptime over a rolling window)
- Usability/accessibility testing (TM-NF-004: responsive UI, contrast, keyboard nav)

Entry Criteria: Stable staging build deployed, synthetic test data seeded, test environment ready.
Exit Criteria: 100% of planned test cases executed, 0 critical defects open, every requirement's acceptance criterion in the SRS RTM satisfied.

5.1 Security Validation

- Verify passwords are stored bcrypt-hashed and never logged or returned in API responses (PRJ-SR-001)
- Verify expired/invalid JWTs are rejected on protected routes with 401 (PRJ-SR-002)
- RBAC checks: a non-member's request to a project/task returns 403 (PRJ-SR-003)
- TLS 1.2+ verification on all client ↔ API traffic (PRJ-SR-004)
- Input fuzzing/injection testing on task/comment fields to confirm sanitization (NoSQL injection, XSS) (PRJ-SR-005)
- Rate-limiting checks on /auth/login and /auth/register

6. Test Environment

Hardware: standard developer laptops/desktops; no special hardware required.
Software: Task Manager staging build (React SPA served separately or via the Express app), Node.js/Express API staging build, MongoDB test database (Atlas free tier or local) seeded with synthetic data.
Tools: Jest/React Testing Library (frontend unit tests), Supertest/Jest (API unit & integration tests), Postman/Newman (manual & scripted API testing), Cypress or Playwright (end-to-end UI tests), JMeter or k6 (TM-NF-001/005 load testing), Jira (the same project tracking scope per SRS §2.5, reused for defect tracking).
Test Data: synthetic user accounts (Admin/Lead/Member roles), sample projects and tasks across all statuses and priorities, sample comments and notifications.

7. Test Schedule

Milestones (indicative — align to the actual course calendar):
- Test case design: 03-Oct-2026
- Environment setup: 04-Oct-2026
- Test execution start: 06-Oct-2026
- Test execution end: 13-Oct-2026
- UAT: 14-Oct-2026 to 16-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports (Jira)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Vismay H S (PES1UG24AM325)	Prepare plan, coordinate execution
Test Engineer	Trisha G Shetty (PES1UG24AM308)	Design & execute test cases, log defects
Developer	Akash Kiran Vernekar (PES1UG24AM338)	Support defect fixes and triage
Product Owner	Basawaraj Panchal (PES1UG24AM350)	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
MongoDB schema drifts as requirements evolve mid-semester (SRS §9 assumes a schema-flexible DB)	Mongoose schema validation plus a short migration note in the PR whenever a schema field changes
Four teammates working on shared modules (auth, task service, board UI) causes merge conflicts or integration breakage	Each Business-layer service (Auth, Task & Project, Notification, Search) is one independently-ownable module; Jira tracks ownership per SRS §2.5
Load-testing 100 concurrent users (TM-NF-005) needs infrastructure the team may not have access to	Use a free-tier load-testing tool (k6 Cloud free tier or local k6/JMeter) against the staging deployment
End-to-end UI tests (Cypress/Playwright) are flaky across contributors' machines	Run E2E suites in CI against a fixed staging environment rather than locally per teammate

11. Assumptions & Dependencies

- A staging deployment of the Node.js/Express API and a MongoDB test database will be available before test execution begins
- Synthetic test data (users, projects, tasks, comments) will be seeded before execution
- The team's existing Jira project (per SRS §2.5) will be reused for defect tracking

12. Suspension & Resumption Criteria

Suspend testing if:
- Staging environment unavailable for >4 hours
- Build is too unstable (blocks >30% of test cases)

Resume testing if:
- Blocking defects are resolved
- Environment is stabilized

13. Test Case Management & Traceability

RTM maps each SRS requirement ID to test cases.
Example:
- TM-F-001 (registration) → TC-Auth-01
- TM-F-002 (login) → TC-Auth-02
- TM-F-010 (create task) → TC-Task-01
- TM-F-012 (task status change) → TC-Task-03
- TM-F-020..023 (assignment & board) → TC-Board-01, TC-Board-02
- TM-F-030 (notifications) → TC-Notif-01

- TM-NF-001 (API response time) → TC-Perf-01
- TM-NF-005 (100 concurrent users) → TC-Perf-02
- PRJ-SR-003 (RBAC) → TC-Sec-03

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage (against the SRS RTM)
- TM-NF-001 API latency (P95)
- TM-NF-002 measured uptime %

Reports:
- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Vismay H S (PES1UG24AM325)	
Dev Lead	Akash Kiran Vernekar (PES1UG24AM338)	
Product Owner	Basawaraj Panchal (PES1UG24AM350)	